Modeling Human Activity States Using Hidden Markov Models

Formative 2 | Machine Learning Operations | African Leadership University

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1. Background and Motivation

Wearable health monitors, fitness trackers, and smart home systems increasingly depend on the ability to recognize physical activity in real time from raw sensor data alone; they do this without depending on explicit user input, and a Hidden Markov Model (HMM) is best suited for problems like this. Because the true activity state is never directly observed; only the continuous, noisy output of accelerometer and gyroscope sensors is available. An HMM explicitly separates the latent activity state from the observable sensor signal and encodes two key properties of human movement: the statistical pattern each activity produces in sensor data (emission probabilities), and realistic constraints on how activities transition over time (transition probabilities). This project implements a complete HMM-based activity recognition pipeline, from data collection through Viterbi decoding and held-out evaluation using smartphone recordings of four activities: Still, Standing, Walking, and Jumping.

2. Data Collection and Preprocessing

Data were collected using an iPhone running Sensor Logger (Chio Tsz Hei), with the Accelerometer (x, y, z in g) and Gyroscope (x, y, z in rad/s) enabled at a fixed 50 Hz sampling rate. The phone was held at waist level for Standing, Walking, and Jumping, and placed flat on a surface for Still. Fifteen takes of approximately 10 seconds each were recorded per activity, yielding ~150 seconds (~2.5 minutes) of labeled data per activity and ~10 minutes total. Each take was exported as a CSV with Unix nanosecond timestamps and per-axis readings, organized into per-activity subdirectories (activity_NN_sensor.csv). As data was collected by a single contributor using one device, no cross-device sampling rate harmonization was required. Four additional unseen test recordings, one per activity, were collected in a separate session and kept strictly isolated from training. Accelerometer and Gyroscope files were merged per take using a nearest-timestamp join on seconds_elapsed, producing a unified six-column table. The merged training dataset contains 30,821 rows across 60 takes (Jumping: 7,766; Standing: 7,726; Walking: 7,689; Still: 7,640). Figure 1 shows raw accelerometer signals for one representative take per activity, illustrating the contrasting motion signatures that motivate the feature selection in Section 3.

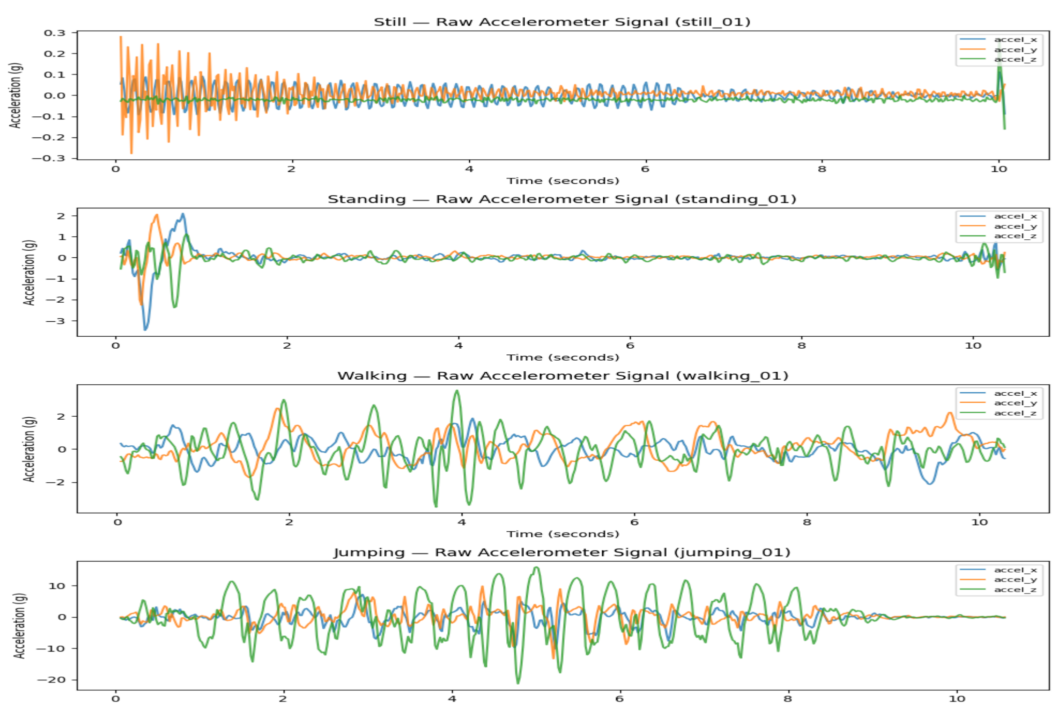


Figure 1. Raw accelerometer signals (accel_x, accel_y, accel_z) for one 10-second take per activity at 50 Hz.

3. HMM Setup and Implementation

3.1 Feature Extraction and Windowing

Each take was segmented into non-overlapping 1-second windows of 50 samples (50 Hz) long enough to capture a full walking stride or jump arc, short enough to avoid spanning activity transitions. This yields 599 training windows (~150 per activity). Sixteen features were extracted per window:

- Time-domain (14): mean and standard deviation per axis for both Accelerometer and Gyroscope (12 features); Signal Magnitude Area (SMA) for Accelerometer and Gyroscope (2 features). Mean captures directional bias; standard deviation quantifies variability; SMA provides a scalar energy measure that increases monotonically from Still to Jumping.
- Frequency-domain (2): dominant frequency and spectral energy, derived from the FFT of the accelerometer magnitude signal after DC removal. Dominant frequency captures the stride cadence of Walking (~1.5–2 Hz) and Jumping (~2–4 Hz), absent in Still and Standing. Spectral energy captures total signal power, highest for Jumping due to its impulsive waveform.

All 16 features were Z-score normalized using statistics fit on training data, applied without refitting to the test set. Normalization is necessary because feature scales span multiple orders of magnitude (standard deviations are single-digit; spectral energy reaches thousands), which would otherwise bias HMM emission calculations toward high-magnitude features regardless of their discriminative value.

3.2 Model Structure, Initialization, and Training

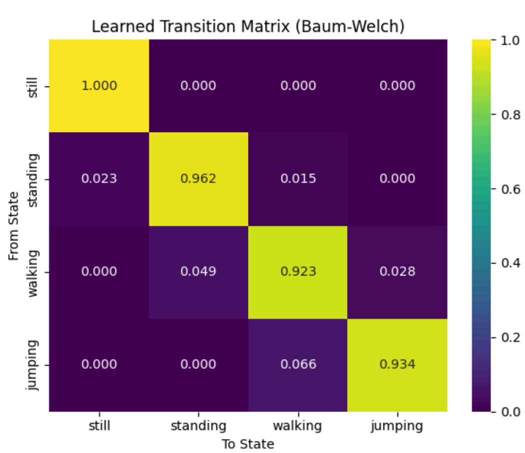
The HMM has four hidden states and uses 16-dimensional diagonal-covariance Gaussian emissions. We initialized the transition matrix A, emission means and variances, and initial probabilities π from labeled training data. We accumulated transition counts from within-take sequences and smoothed them with an additive factor of 0.1. We estimated emission parameters as the average mean and variance per class, using epsilon 1e-6 for stability. This supervised initialization helps avoid the poor local optima that often occur with random initialization in small datasets.

The Viterbi algorithm works in log-space, using a log-delta matrix and psi backpointers to prevent underflow. It achieved a mean training accuracy of 0.973, with a range from 0.900 to 1.000. We ran the Baum-Welch EM refinement with a tolerance of 1e-3 and a maximum of 50 iterations. In the E-step, we stored gamma values to reuse in the M-step variance update for consistent parameters. The training converged in 7 iterations, with log-likelihood increasing from 7,534 to 9,444. This shows genuine convergence rather than just hitting an iteration limit. Post-refinement accuracy is 0.962. In independent validation, hmmlearn's GaussianHMM with random initialization achieved a mapped accuracy of 0.733 under the same settings. This confirms that supervised initialization significantly outperforms random initialization with this dataset.

4. Results and Interpretation

4.1 Transition Matrix and Decoded Sequences

Figure 2 shows the learned transition matrix. The strong diagonal (0.923–1.000) reflects activity persistence within 1-second windows; Baum-Welch introduced plausible off-diagonal transitions (Standing→Walking: 0.015; Walking→Jumping: 0.028; Jumping→Walking: 0.066). Figures 3 and 4 show Viterbi-decoded training sequences: walking_01 is perfectly decoded across all 10 windows; still_08 shows a transient misclassification of Still as Standing in windows 0–2, attributable to residual vibration as the phone was placed onto the recording surface.



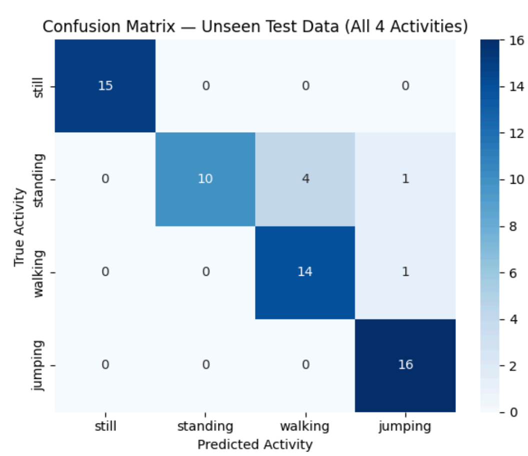


Figure 2 (left). Learned transition matrix. Figure 6 (right). Confusion matrix on unseen test data.

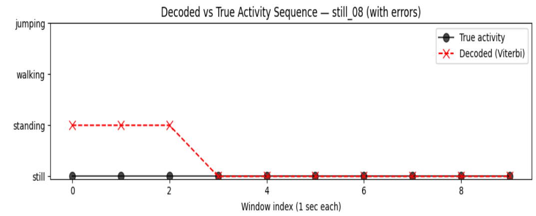


Figure 3 (left). Decoded vs. true sequence for walking_01 — all windows correct. Figure 4 (right). still_08 — transient error windows 0–2.

4.2 Emission Distributions

Figure 5 shows the learned emission means per hidden state. Jumping is strongly separated by high variance, SMA, and spectral energy (normalized values 1.4–1.6). Still and Standing share near-identical low-energy profiles, explaining their occasional confusion. Walking occupies an intermediate position. The dominant_freq feature distinguishes Still (1.46) from Walking (−0.64) despite their similar raw energy levels, demonstrating the value of frequency-domain features.

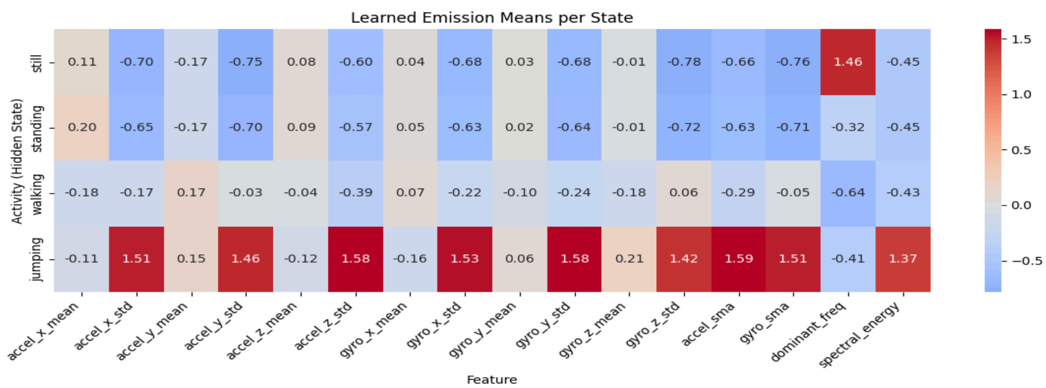


Figure 5. Learned emission means per hidden state (normalized Z-scores). Jumping is strongly differentiated; Still and Standing share overlapping low-energy profiles.

4.3 Evaluation on Unseen Data

The model was evaluated on four unseen test recordings collected in a separate session (one per activity, never seen during training). Overall test accuracy is 0.902. Table 1 summarizes per-activity metrics. Still and Jumping achieve perfect sensitivity (1.000); Walking achieves 0.933 (one window misclassified as Jumping); Standing is the most challenging at 0.667, with four windows predicted as Walking and one as Jumping — consistent with the overlapping emission profiles in Figure 5. Figure 6 (above, right) shows the full confusion matrix.

Table 1. Per-activity evaluation metrics on unseen test data (overall accuracy: 0.902).

Activity	N Samples	Sensitivity	Specificity	Accuracy
Still	15	1.000	1.000	1.000
Standing	15	0.667	1.000	0.918
Walking	15	0.933	0.913	0.918
Jumping	16	1.000	0.956	0.967

5. Discussion and Conclusion

The HMM achieves 96.2% accuracy on training data and 90.2% on unseen test data. This shows that a small dataset collected from smartphones is enough to train a functional activity classifier. Jumping and Still are clearly separable at the two ends of the feature space. Walking stands out due to its rhythmic frequency and intermediate SMA. Standing is the main source of errors, as its feature profile overlaps with both Still at rest and Walking during slight postural sway. This overlap is visible in both Figure 5 and Figure 6. The transition matrix reflects realistic movement dynamics, even though it was trained on single-activity recordings. Residual vibration during the placement of Still recordings leads to the main transient misclassification, as seen in still_08. Future improvements could include using frequency features from the gyroscope, full covariance Gaussian emissions, continuous multi-activity training sequences, and an onset-exclusion step for Still recordings. The resultsdemonstrate that a principled probabilistic sequence model, carefully initialized on labeled sensor data, can achieve robust activity recognition with minimal data collection overhead.

Appendix: Data Collection Summary

Contributor	Device	Sampling Rate	Takes / Activity
Glory Paul	iPhone (iOS)	50 Hz	15